

Library Management System - Capstone

Full Stack Development Project

User Manual

Overview

- LMS allows members to borrow books and make reservations of books not available
- They can also manage their book loans and reservations with the systems

LMS Membership

- ❑ Only members are allowed to borrow books, they need to create an account with email to be a member
- ❑ Membership lasts for 1 year after account creation after which they need to request for renewal.
- ❑ Members cannot borrow any books if their membership is suspended or expired

Book Loan Rules

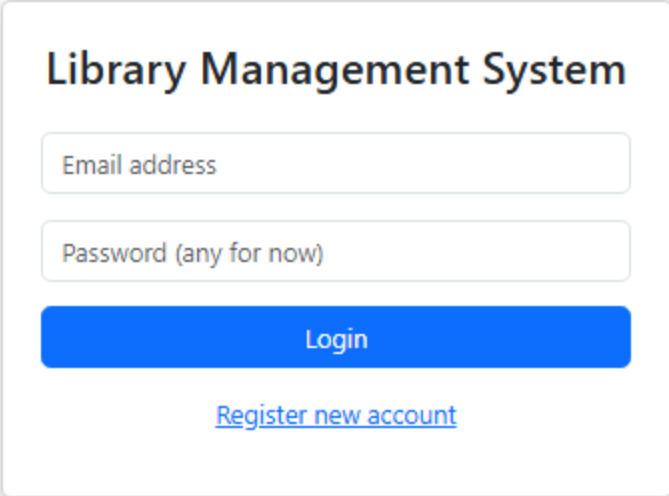
- Members can have up to 3 active loans
- Date due for each book is 14 days
- Renewal can be requested up to 2 more times; each renew will move the due date forward by 14 days
- Fines are calculated per book at \$0.50 one day after date due onwards, up to a maximum of \$20 per book.
- If a member has cumulative fines of \$10 or more, borrowing or reservations are disabled until fines are paid up

Book Reservation Rules

- Members can reserve books already borrowed, up to 3 copies
- Reservation has a due date of 14 days
- Only the member who reserved the book can borrow it after its been returned.
- If the reservation is canceled or expired (automatic cancel), the book becomes available again.
- Same rules of a maximum of 3 book loans apply.

Login to LMS

- Upon launch the web page, you will be presented by a login page
- If you have a valid account, you can login.
- If you need an account, contact the Librarian.



The image shows a login form for a 'Library Management System'. The form is white with rounded corners and a subtle shadow, set against a light gray background. It contains two input fields: 'Email address' and 'Password (any for now)'. Below these fields is a prominent blue 'Login' button. At the bottom of the form, there is a blue hyperlink that reads 'Register new account'.

Library Management System

Email address

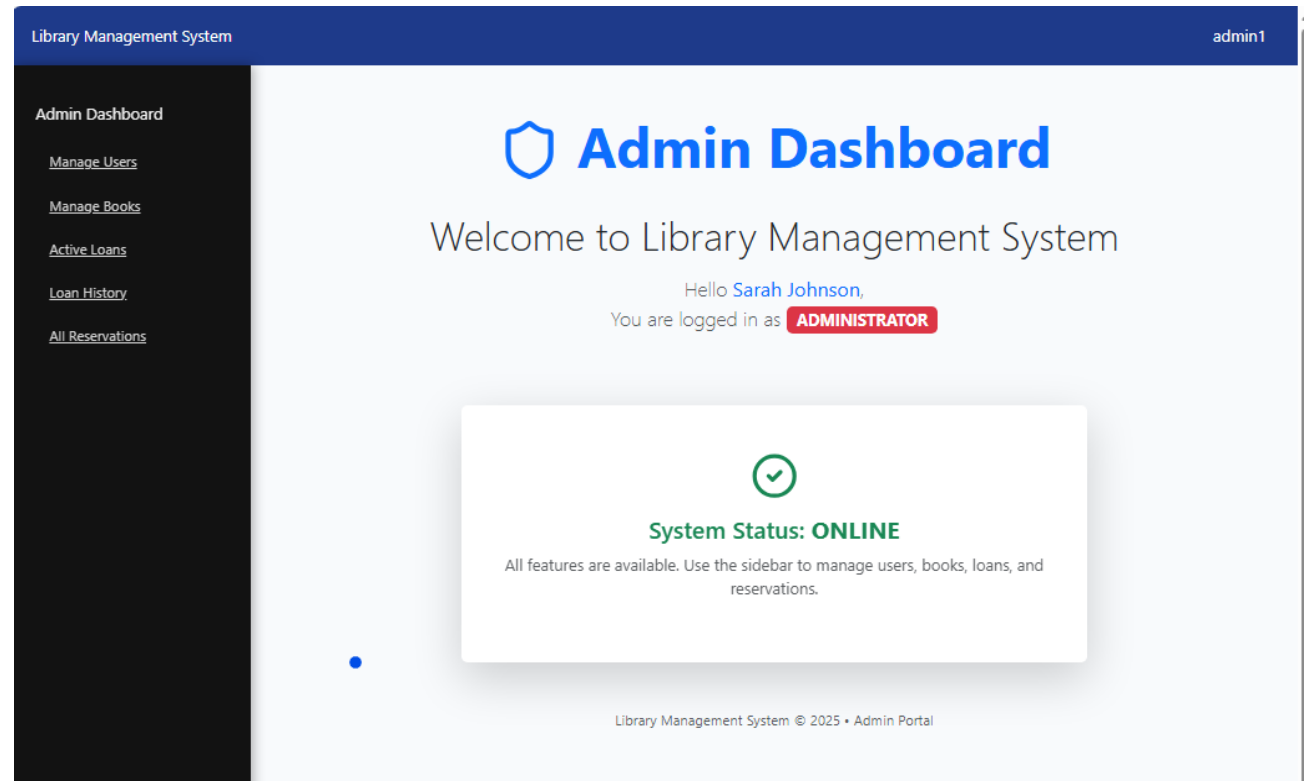
Password (any for now)

Login

[Register new account](#)

Admin Dashboard

- This will show you the system status of the LMS
- The main menu is on the left
- To logout, click the user menu on the top right hand corner



My Profile

- You can launch your Profile page from the user menu
- This allows you to update some of your own person information and change your password

The screenshot displays a web application interface. At the top, a dark blue header bar contains the text 'ent System' in large blue letters. To the right of this header, a user menu is visible, showing 'user8' and a dropdown menu with 'Profile' and 'Logout' options. Below the header, the main content area features a 'My Profile' section. This section has a blue header with the title 'My Profile' and an 'Edit' button. The profile form contains the following fields and values:

Full Name *	Sarah Johnson
Email	admin1@library.com
Phone	1123456789
Address	123 Main St, Admin City
Account Type	Admin

At the bottom of the profile form, there is a 'Change Password' button.

Manage User

- You can add, update, view and delete users from here
- For members, you can update their status if needed
- You can also view the members status from there

Manage Users

[+ Add User](#)

Email	Name	Role	Member ID	Status	Actions		
admin1@library.com	Sarah Johnson	Admin	—	—	View	Edit	Delete
admin2@library.com	Michael Brown	Admin	—	—	View	Edit	Delete
alice.wong@gmail.com	Alice Wong	Member	MEM-0003	Active	View	Edit	Delete
bob.smith@yahoo.com	Bob Smith	Member	MEM-0004	Active	View	Edit	Delete
carla.martinez@outlook.com	Carla Martínez Coda	Member	MEM-0005	Active	View	Edit	Delete
david.lee@student.edu	David Lee	Member	MEM-0006	Active	View	Edit	Delete
emma.davis@hotmail.com	Emma Davis	Member	MEM-0007	Active	View	Edit	Delete
admin@library.com	Kelvin Wong	Admin	—	—	View	Edit	Delete
user1@email.com	John Doe User1	Member	MEM-0012	Active	View	Edit	Delete
user2@email.com	Kelvin Wong	Member	MEM-0019	Active	View	Edit	Delete
user3@email.com	Victory Kang User3	Member	MEM-0021	Active	View	Edit	Delete
user4@email.com	Peter Parker User4	Member	MEM-0022	Active	View	Edit	Delete

Manage Books

- You can add, update and delete books from here
- You can use the search box to narrow your search by ISBN, title, author, category, publisher and published years.
- You can see how many available copies and increase or decrease the total copies for a book.
- You cannot decrease the total copies less than the number on loan.

Manage Books

[+ Add Book](#)

Search By

Title

Search Term

Enter title...

Query

Clear

Introduction to Algorithms
ISBN: 9780262033848
Author: Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein
Publisher: MIT Press
Year: 2009
Category: Computer Science
Copies: 0 / 1

EditDelete

Structure and Interpretation of Computer Programs
ISBN: 9780262510875
Author: Harold Abelson, Gerald Jay Sussman
Publisher: MIT Press
Year: 1996
Category: Programming
Copies: 1 / 1

EditDelete

Clean Code: A Handbook of Agile Software Craftsmanship
ISBN: 9780134685991
Author: Robert C. Martin
Publisher: Prentice Hall
Year: 2008
Category: Software Development
Copies: 0 / 1

EditDelete

Code: The Hidden Language of Computer Hardware and Software
ISBN: 9780735611313
Author: Charles Petzold
Publisher: Microsoft Press
Year: 2000
Category: Computer Science
Copies: 1 / 1

EditDelete

The Mythical Man-Month: Essays on Software Engineering
ISBN: 9780201835953
Author: Frederick P. Brooks Jr.
Publisher: Addison-Wesley
Year: 1995
Category: Software Engineering
Copies: 0 / 1

EditDelete

Design Patterns: Elements of Reusable Object-Oriented Software
ISBN: 9780201633610
Author: Erich Gamma, Richard Helm, Ralph Johnson, John Vlissides
Publisher: Addison-Wesley
Year: 1994
Category: Software Design
Copies: 1 / 1

EditDelete

Active Loans

- Allows you to view and manage active loan status of all members
- You can return and renew books on behalf of a member, with the same rules applying regarding renew.
- You can also view cumulated fines for a member and the amount

📖 Active Loans (Admin View)

Search By

All Fields

▼

Search Term

Enter search term...

☐ Show Only With Fines

MEM-0021 - Victory Kang User3						\$ Fine: \$20.00
ISBN	Title	Borrowed	Due	Fine	Renewals	Actions
9781449373320	Designing Data-Intensive Applications	02 Aug 2025	16 Aug 2025 (Overdue)	\$20.00	0/2	
9780262033848	Introduction to Algorithms	09 Dec 2025	23 Dec 2025	\$0.00	0/2	

MEM-0026 - Mary Brown							
ISBN	Title	Borrowed	Due	Fine	Renewals	Actions	
9780134685991	Clean Code: A Handbook of Agile Software Craftsmanship	10 Dec 2025	24 Dec 2025	\$0.00	0/2		
9780201657883	Programming Pearls	10 Dec 2025	24 Dec 2025	\$0.00	0/2		
9780131103627	The C Programming Language	10 Dec 2025	24 Dec 2025	\$0.00	0/2		

MEM-0022 - Peter Parker User4						\$ Fine: \$3.50
ISBN	Title	Borrowed	Due	Fine	Renewals	Actions
9780451524935	1984	20 Nov 2025	04 Dec 2025 (Overdue)	\$3.50	0/2	

MEM-0027 - Philip Starck

Loan History

- This will show you the history of all book loans and their duration

🕒 All Loans History (Admin View)

Search By		Search Term					
All Fields		Search by member, book title, ISBN...					
Member ID	Name	ISBN	Title	Borrowed	Returned	Renewals	Duration
MEM-0012	John Doe User1	9780262033848	Introduction to Algorithms	06 Dec 2025	06 Dec 2025 20:36	0/2	1 day
MEM-0012	John Doe User1	9780262033848	Introduction to Algorithms	06 Dec 2025	06 Dec 2025 21:47	0/2	1 day
MEM-0012	John Doe User1	9780262510875	Structure and Interpretation of Computer Programs	06 Dec 2025	10 Dec 2025 10:52	0/2	4 days
MEM-0012	John Doe User1	9780134685991	Clean Code: A Handbook of Agile Software Craftsmanship	20 Nov 2025	06 Dec 2025 21:48	0/2	17 days
MEM-0012	John Doe User1	9780262033848	Introduction to Algorithms	06 Dec 2025	06 Dec 2025 22:29	0/2	1 day
MEM-0021	Victory Kang User3	9780201633610	Design Patterns: Elements of Reusable Object-Oriented Software	07 Dec 2025	09 Dec 2025 00:33	0/2	2 days
MEM-0022	Peter Parker User4	9780201835953	The Mythical Man-Month: Essays on Software Engineering	20 Nov 2025	09 Dec 2025 00:59	0/2	19 days
MEM-0022	Peter Parker User4	9780201896831	The Art of Computer Programming, Volume 1: Fundamental Algorithms	20 Nov 2025	10 Dec 2025 10:39	0/2	21 days
MEM-0022	Peter Parker User4	9780735611313	Code: The Hidden Language of Computer Hardware and Software	20 Nov 2025	10 Dec 2025 10:31	0/2	21 days

All Reservations

- Here you can manage reservations of members.
- You can cancel their reservation or set them to fulfilled.
- You can view reservations and their status like expiration from here.

Manage All Reservations (Admin)

Search By

All Fields

▼

Search Term

Search reservations...

Clear

Member ID	Name	ISBN	Title	Reserved	Expires	Status	Actions
MEM-0019	Kelvin Wong	9780262510875	Structure and Interpretation of Computer Programs	06 Dec 2025 23:45	20 Dec 2025	Active	
MEM-0022	Peter Parker User4	9780201835953	The Mythical Man-Month: Essays on Software Engineering	10 Dec 2025 23:47	24 Dec 2025	Active	
2 Total		2 Active	0 Expired	0 Fulfilled			

Login notifications

- When a member logs in, the system checks for the following
 - If their status is Suspended, they are not allowed to log in; they must contact the Librarian
 - If their status is Expired, they can only login if they renew your membership
 - If their membership expiration date is 60 days or less, the system will ask them to renew membership. If they choose to renew later, they will still be allowed to log in.
 - If they have any loans are the past due date, they will get a notification upon logging into the system

END